Akhilesh Pingle

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SUMMARY

AI/ML Engineer with 3+ years of experience developing production AI solutions across Conversational AI, Retrieval-Augmented Generation (RAG), recommendation systems, and demand forecasting. Built enterprise LLM applications using LangChain, LangGraph, Hugging Face Transformers, and FastAPI, while developing machine learning pipelines with PyTorch, Scikit-learn, PySpark, and Databricks. Experienced in deploying real-time AI inference services on AWS, optimizing model performance, and delivering scalable, cloud-native AI applications.

TECHNICAL SKILLS

Artificial Intelligence: AI Agents, Conversational AI, Semantic Search, Contextual Retrieval, Real-Time Inference
Generative AI & NLP: Large Language Models (LLMs), Generative AI, LangChain, LangGraph, LangSmith, Retrieval-Augmented Generation (RAG), Prompt Engineering, Hugging Face Transformers, NLP, Vector Databases (FAISS, Pinecone)
Machine Learning: Machine Learning, Deep Learning, Supervised Learning, Scikit-learn, PyTorch, Recommendation Systems, Forecasting Models
Data & AI Infrastructure: Python, SQL, Apache Spark, PySpark, Databricks, Airflow, ETL Pipelines
Cloud, MLOps & Deployment: AWS (SageMaker, Lambda, EKS), Docker, Kubernetes, GitHub Actions, CI/CD
Backend Engineering & Observability: FastAPI, REST APIs, Microservices, PostgreSQL, Redis, Linux, Git

PROFESSIONAL EXPERIENCE

AI Engineer, *Moveworks* Jan 2025 - Present | CA, USA

- Built Conversational AI workflows using LangChain and FastAPI to automate enterprise support requests, reducing repetitive ticket escalations by 34% across enterprise support operations.
- Integrated Retrieval-Augmented Generation (RAG) pipelines with Pinecone and Semantic Search capabilities, enabling contextual responses from enterprise knowledge sources for conversational assistance systems.
- Improved Real-Time Inference stability by deploying containerized AI services on AWS EKS with Kubernetes orchestration and Redis caching, reducing response latency by 26% during peak workloads.
- Monitored prompt execution and response traces for enterprise LLM workflows through LangSmith, improving debugging visibility and response validation for conversational AI systems.
- Automated multilingual document understanding workflows using Hugging Face Transformers, Python microservices, and PostgreSQL storage layers for enterprise assistance platforms handling internal policy retrieval.

ML Engineer, *Naykaa* Sep 2021 - Jul 2023 | PUNE, India

- Built demand forecasting models using Scikit-learn and historical sales, promotional, and seasonal datasets, improving inventory planning accuracy by 21% while reducing stock imbalance across product categories.
- Developed personalized recommendation pipelines using PyTorch, feature engineering, and customer behavior signals, improving product ranking relevance and increasing recommendation engagement across digital commerce experiences.
- Engineered distributed data processing workflows with PySpark and Databricks to transform large-scale transaction and catalog datasets, reducing reporting inconsistencies by 17% for merchandising analytics teams.
- Automated feature engineering and ETL pipelines using Python, Airflow, and SQL, ensuring reliable training datasets for recommendation and demand forecasting models across evolving retail inventories.
- Evaluated machine learning models through feature importance analysis, cross-validation, and performance monitoring, improving recommendation consistency by 19% while supporting continuous model refinement.
- Deployed forecasting models on AWS SageMaker through REST APIs, enabling scalable real-time inference for inventory planning applications supporting pricing and demand optimization initiatives.

CERTIFICATION

- **AWS Certified Solutions Architect – Associate (SAA-C03) – (Link)**

PROJECTS

Agentic Document Intelligence Platform | Python, LangChain, LangGraph, AWS Bedrock, FAISS, FastAPI, PostgreSQL

- Developed a Retrieval-Augmented Generation (RAG) platform that indexed invoices and contracts using FAISS vector embeddings, delivering contextual document retrieval through FastAPI APIs deployed on AWS.
- Engineered an agentic workflow with LangGraph and tool-calling capabilities to orchestrate document analysis, validation, and response generation, reducing manual review effort by 60%.
- Implemented prompt engineering, contextual retrieval, and response evaluation to improve answer quality while integrating PostgreSQL for metadata management and audit tracking.

Serverless Sales Forecasting API | Python, Scikit-learn, AWS Lambda, API Gateway, DynamoDB, GitHub Actions

- Built a serverless machine learning inference API using AWS Lambda and API Gateway to deliver real-time sales forecasts with DynamoDB for low-latency data persistence.
- Developed forecasting models with Scikit-learn and automated model deployment using GitHub Actions CI/CD, enabling reliable daily releases with minimal operational overhead.
- Optimized inference performance and API scalability through event-driven architecture, reducing infrastructure costs while supporting variable production workloads.

EDUCATION

Master of Science in Computer Science, *California State University* Aug 2023 - May 2025 | CA, USA
Bachelors of Science in Information Technology, *Savitribai Phule Pune University* Jun 2019 - May 2023 | MH, India